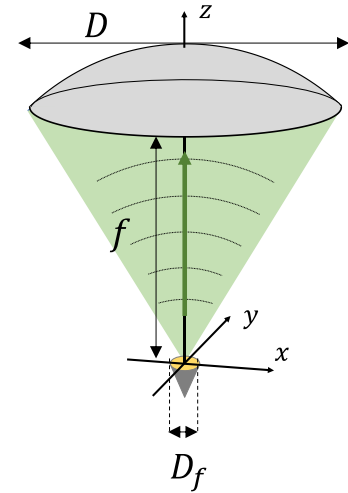


Assignment 2: Reflector Antennas

Write a MATLAB routine to analyze a parabolic reflector antenna. Suppose that the reflector is illuminated by a circular antenna feed with an electric equivalent current which is y-polarized with a unitary and uniform amplitude distribution. The operating frequency is 500 GHz. The aperture of the circular antenna feed has a diameter of $D_f = 4\lambda$. Suppose that the parabola has a focal length of $f = 5$ m.



Provide the following results in your report:

1. **(1 point)** Provide the normalized far field patterns of the circular antenna feed at the two main cuts (E- and H- planes) in dB. Explain what governs the shape of this far field and its variation over the azimuth planes.
2. **(2 point)** Plot the amplitude of the equivalent aperture current distribution for a reflector with $f/D = 2$ in dB. Plot both the magnetic and electric equivalent currents in 2D maps (e.g. a u-v map). Separate these plots into two, one for the x- and one for the y-polarization.
3. **(2 point)** Plot the normalized far field patterns radiated by the parabolic reflector for $f/D = 1$ at the two main cuts (E- and H- planes) in dB. Compare this far field to the one where the equivalent aperture of the reflector has a uniform current distribution, i.e. compare the far field to the corresponding Airy pattern.
4. **(3 point)** Plot the taper, spillover, and aperture efficiencies vs the diameter D of the reflector for $0.6 < f/D < 6$ in linear scale. Explain how different f/D values influenced these efficiencies. What criteria leads to the maximum aperture efficiency in this plot?
5. **(2 point)** Plot the maximum possible directivity (i.e. for a uniform aperture distribution), directivity and gain of the reflector antenna vs the diameter D for $0.6 < f/D < 6$ in dBi. Explain how different f/D values influenced these parameters.

Assignment's report should contain your comments on the obtained results. Providing only the requested plots does not count as a complete report.